

Maxwell Hanks

Brisbane, QLD

Email ◊ LinkedIn ◊ Github ◊ Portfolio

EDUCATION

The University of Queensland (UQ)

2026 - Present

Bachelor of Mathematics (Mathematical Artificial Intelligence/Computer Science)

Cum. GPA: 6.25 / 7.0 (WAM: 83)

Relevant Coursework: Algorithms & Data Structures, Deep Learning, Object-Oriented Programming, Statistical Inference

Queensland University of Technology (QUT)

2025 (Withdrew)

Bachelor of Engineering (Honours)/Bachelor of Mathematics

Major in Computer & Software Systems Engineering, Statistics

EXPERIENCE

Workshop Presenter

2026 - Current

Connect Education

Brisbane, QLD

- Delivered targeted exam-preparation workshops, using IA/EA data to tailor content and close knowledge gaps, guided students through past exam questions to build strong exam technique and align responses with QCAA criteria.

Academic Tutor

2025 - Current

Private, All Access Education, KIS Academics

Hybrid

- Completed 200+ hours of tutoring for high school students across various Australian curriculum, delivering targeted support in study skills, exam preparation, and assignment planning.

PROJECTS

Rethinking ImageNet Pretraining for Label-Efficient Satellite Image Classification — *Python*

2026

Compared random, ImageNet-supervised, and satellite self-supervised ResNet-50 initializations across linear probing and fine-tuning regimes, measuring label efficiency on EuroSAT and cross-sensor generalization on out-of-domain satellite datasets. Derived practical guidelines for pretraining strategy selection in label-scarce Earth observation settings.

Continuous-Time Optimal Execution via HJB & Neural Networks — *Python, MATLAB*

2026

Formulated optimal trade execution as a continuous-time stochastic control problem, deriving the HJB equation for liquidation. Solved the linear-quadratic benchmark case analytically and extended to nonlinear market impact models using neural network approximation, with ε -optimality validation against dynamic programming.

Multi-Factor Equity Risk Model (Barra-style) — *Python, NumPy, Pandas*

2026

Built an institutional-grade fundamental factor risk model over a 94-stock US universe: engineered six point-in-time style factors, estimated daily factor returns via cross-sectional WLS (1,200+ regressions), and constructed the asset covariance matrix using EWMA/Newey-West factor covariance with PSD enforcement and shrinkage-stabilized specific risk. Implemented Euler risk decomposition and ex-ante tracking error for portfolio analysis, and validated volatility forecasts out-of-sample via walk-forward bias statistics.

EXTRA-CURRICULAR & LEADERSHIP

Academic/Leadership: SSLG Representative for STAT3007, UQ Science Leader Development Program 2026 (pathway to UQ Science Academy).

Member of: UQPAIN, UQCS (Computing), UQCS (Cyber Squad), UQ Fintech, UQMSS, UQWS

Competitions: IMC AlgoJam 3 — *3rd of 124*, Jane Street Inter-University Mathematics Competition, Susquehanna Algothon '26 — *Top 200*

Activities: Citadel Securities Cryptography Challenge, IMC Emerging Talent Function, Jane Street Dinner, Optiver Volatility Lab, Susquehanna/UQ FinTech Poker Night

TECHNICAL SKILLS

Languages Python, C/C++, HTML/CSS, Java, Lua, MATLAB, R, TypeScript

Libraries pandas, NumPy, PyTorch, scikit-learn, Matplotlib, statsmodels, Seaborn, Tensorflow

Tools Excel, Git/GitHub, Jax, Jupyter, LaTeX, Linux/Unix, Vim